

5/H-73 (vi) (c) (Syllabus-2015)

2 0 2 4

(November–December)

COMPUTER SCIENCE

(Honours)

(Object-Oriented Programming Through Java)

(CS-502 CT)

Marks : 38

Time : 2 hours

*The figures in the margin indicate full marks
for the questions*

Answer one question from each Unit

UNIT—I

1. (a) What is narrowing and widening in data type conversion? Explain with a suitable example. 3+1=4
- (b) Find the output of the following program segment : 2

```
int a = 6;
int b = a++;
System.out.println(a);
System.out.println(b);
a = 6;
b = ++a;
System.out.println(a);
System.out.println(b);
```

(2)

- (c) List any two concepts that are considered as the foundation of Object-Oriented Programming paradigm. 2
2. (a) Explain how you can iterate through an array using (i) for loop and (ii) enhanced for loop with an example. 2
- (b) Create a class named "point" with two private data members— x and y , of type integer. Write a no-arguments constructor that assigns values 1 to both x and y . Write an overloaded constructor with two integer arguments that assigns values to x and y . Write public methods, $getX$ and $getY$ to read values of X and Y respectively. Finally, test all constructors and methods in a Tester class. $2+2+2=6$

UNIT—II

3. (a) What is the difference between the 'extends' keyword and the 'implements' keyword? 2
- (b) Explain the meaning of the following keywords 'try', 'catch', 'throw', 'finally' and 'throws', with a suitable example. $5\frac{1}{2}$
4. (a) Can a class be a superclass and a subclass at the same time? Explain with a suitable example. $\frac{1}{2}+2=2\frac{1}{2}$

(3)

- (b) Explain, with a suitable example, how the runnable interface can be used to create a new thread in Java. 5

UNIT—III

5. (a) How does a generic class implement type safety? How can you create a generic method and invoke it? $1\frac{1}{2}+3=4\frac{1}{2}$
- (b) Write the code to use the file class in Java to display the following : 3
- (i) Create a file, given the filename
- (ii) Display the size of the file in bytes
- (iii) Display the date the file was last modified
6. (a) Let the string $s1$ be "Welcome" and $s2$ be "welcome". Write the code to replace all occurrences of the character 'e' with 'E' in $s1$ and assign the new string to $s2$. 2
- (b) Create an ArrayList of integers and add few elements to it. Next convert this ArrayList to an array of integers. $2+1\frac{1}{2}=3\frac{1}{2}$
- (c) Distinguish between character streams and byte streams. 2

UNIT—IV

7. (a) How do you create and embed a Java Applet in an HTML page? Provide a

simple example of an applet that displays a message and explain the necessary HTML code. $2+2=4$

- (b) What are anonymous inner classes and how are they used in event handling? $3\frac{1}{2}$
8. (a) Explain how event listeners are used in Java's event delegation model. Provide a suitable example showing how the ActionListener interface can respond to a Button click event. 5
- (b) How does an applet differ from a Java application? $2\frac{1}{2}$

UNIT—V

9. (a) Explain how to create a simple client-server application in Java using TCP/IP sockets. Provide code for both the client and the server, and explain how they communicate with each other. $2+2\frac{1}{2}+1\frac{1}{2}=6$
- (b) Explain the life cycle of a servlet. $1\frac{1}{2}$
10. (a) Explain the steps involved in establishing a JDBC connection to a database. Provide code to explain each step. 6
- (b) Discuss the advantages of servlets. $1\frac{1}{2}$

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